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The Journal of Sports Medicine and Physical Fitness 2024 Jun 11
DOI: 10.23736/S0022-4707.24.15900-2

REVIEW

EXERCISE PHYSIOLOGY AND BIOMECHANICS

Injury prevention in professional soccer players: can re-warmup training be organized in the post-warmup and half-time during a soccer game in a short time?

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A B S T R A C T

INTRODUCTION: Passive time, during the post-warmup and half-time, decreases athletes' performance and increases injury risk factors in the active phases of the soccer match. Objective. This narrative review aims to research and synthesize existing evidence to identify brief re-warmup strategies that may find applicability in the post-warmup and half-time of a soccer match.
EVIDENCE ACQUISITION: The analysis was conducted on PubMed, Web of Science, PEDro, SPORTDiscus and Google Scholar. Due to the lack of evidence, no temporal time was established, preferring most up-to-date articles. The data were synthesized in relation to the objectives, following the Preferred Reporting Items for Systematic Review and Meta-Analysis (PRISMA) guidelines.
EVIDENCE SYNTHESIS: Our research yielded a total of 753 studies, 463 related to re-warmup, 136 articles on re-warmup in soccer, and 154 articles on re-warmup during half-time. Of these, 26 met the inclusion objectives and were included in this research. Our findings confirm that a re-warmup can mitigate the decremental effect of static rest on performance. From the studies considered, we have identified a work that emphasizes how a 1-minute warmup of high-intensity exercise at speed corresponding to 90% of VO_{2max} can prevent decreases related to passive time, in sprint performance and muscle strength, as well as improve muscle temperature. These results could apply to both post-warmup and half-time scenarios of a soccer match.
CONCLUSIONS: From this narrative review, it has been possible to highlight a one-minute high-intensity re-warmup that improved sprint performance, increased core temperature, and enhanced muscle activation, not leading to additional physiological or psychological fatigue.

(*Cite this article as:* D'Onofrio R, Laterza F, Sannicandro I, Masucci M, Bovenzi A, Manzi V. Injury prevention in professional soccer players: can re-warmup training be organized in the post-warmup and half-time during a soccer game in a short time? J Sports Med Phys Fitness 2024 Jun 11. DOI: 10.23736/S0022-4707.24.15900-2)

KEY WORDS: Warm-up exercise; Soccer; Athletic injuries.

Introduction

Soccer is one of the sports characterized by its intermittent nature, which imposes a significant demand on players throughout a 90-minute match. They experience approximately 1200-1400 changes in activity and engage in 150-200 short, high-intensity efforts lasting 1-6 seconds.[1] In the training sessions of elite players, the intensity is set at 85-90% of their maximum heart rate, with an average oxygen uptake (VO_2) of 70-80% of the maximum.[1] Despite the relatively low average match speed (110-140 m/min) observed during a 90-minute match, soccer players can still experience substantial neuromuscular fatigue and energy expenditure.[1]

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Possible risk factors for injury during a warmup

The qualitative expression of movement[2] and the reduction of risk factors for re-injury are positively correlated with all game skills, creating a coherent and harmonious flow between technical and athletic movements that aim for optimal performance and minimize the risk of injury and re-injury.[2]

Most athletes demonstrate a dominant limb in their technical gestures, which can lead to repetitive movements and contribute to muscle imbalances in terms of strength and flexibility. These imbalances, in turn, are associated with asymmetrical posture[3] and structural instability, significantly increasing the risk factors for injury and re-injury.[2, 3] Specific soccer movements such as cutting maneuvers, changes of direction, sprints, and jumps are known to be linked to moments of vulnerability that further enhance the risk factors for injuries. Conducting a thorough postural assessment specific to the sport can aid in identifying risk factors for traumatic injuries, which often occur during tactical situations such as ball recapture in soccer.

Similar to other intermittent sports, soccer involves an initial pre-match warmup followed by a 10-15-minute break before the start of the match. The game consists of two consecutive periods of play, each lasting 45 minutes, with a 10- to 20-minute half-time break in between. However, the presence of passive time during these breaks negatively impacts the athletes' performance and increases the risk of injury during the match.[4, 5]

Arliani[6] recently conducted a study that revealed when the majority of injuries occurred in soccer matches. They found that 23.6% of injuries took place in the last 15 minutes of the first half, 21.7% in the central third of the second half, 9.4% within the first 15 minutes of the match, and 13.4% following the half-time break.

Muscular injuries account for a significant portion of the overall injuries in soccer. In the FIFA World Cup Injury Qatar 2022, a total of 54 injuries were reported, with 15 of them (27.7%) classified as muscular injuries (Figure 1).[7]

Re-warmup: where are we going?

A recent study by González-Devesa *et al.*[8] highlights the negative impact of passive rest on athletes' physiological functions, while emphasizing the potential benefits of re-warmup activities in improving athletes' conditioning and sports performance.[9] One challenge in implementing re-warmup strategies is the limited time available during halftime, which is often dedicated to tactical instructions and feedback from the coach. This can pose a significant obstacle to incorporating re-warmup activities into the halftime routine. Currently, the relationship between scientific evidence and practical applicability in this context remains poorly understood. Therefore, we conducted a narrative review aimed to synthesize existing evidence and identifying practical re-warmup strategies that can be effectively implemented during the post-warmup period (*i.e.*, the time between the end of the warmup and the beginning of the match) and halftime (*i.e.*, the break between the first and second half of the match).

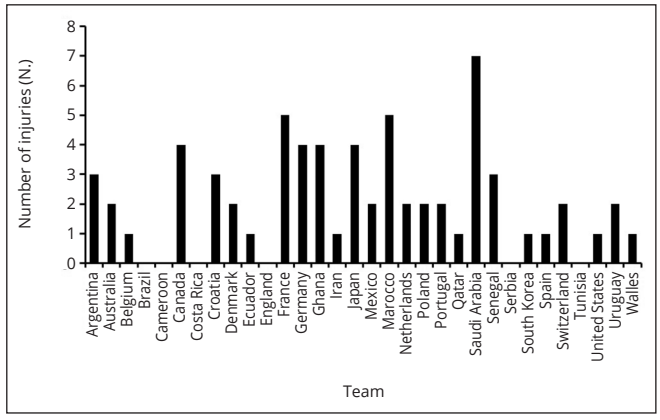

Figure 1.—Total injuries by team during the FIFA World Cup Injury Qatar 2022.

Evidence acquisition

This study aimed to report the literature on brief re-warmup strategies that can be used with professional soccer athletes. Given the aim of this study, the authors decided to draft a review following five steps:

- identifying a research question: is a re-warmup strategy that lasts less than 1,30 min reported in the literature? Can professional soccer athletes use it during the post-warmup and the half-time break?
- identification of the studies: no temporal time limit was established due to the lack of evidence; however, most up-to-date articles were preferred. The search was conducted on PubMed, Web of Science, PEDro, SPORTDiscus, and Google Scholar. The so-called gray literature (*i.e.*, unpublished lectures, abstracts, theses, and reports) was not considered. For indexing, we used MeSH (Medical Subject Headings) Terms. The collected data were synthesized by the first author in accordance with the Preferred Reporting Items for Systematic Review and Meta-Analysis (PRISMA) guidelines;
- selection of the studies: the articles on the re-warmup

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in soccer were considered. All the articles were evaluated based on the relevance and applicability of their results. References from the articles selected were checked for a possible presence of articles important for the aim of this investigation. The first author's opinion was crucial when doubts about the inclusion or exclusion of articles were raised;

• data extraction: the screening process does not provide for a priori eligible criteria in the narrative review. Therefore, the authors used the method proposed by Baethge[10] to identify the most relevant articles. The first author extracted the data, which was then overseen by the other authors;

• results of studies selected: Morgan's methodology[11] was applied to analyze and synthesize the contents.

This review was focused on the narrative description of the main results obtained by the studies included in this investigation to identify brief re-warmup strategies applicable to professional soccer athletes.

Evidence synthesis

The databases analyzed yielded a total of 753 articles, 463 related to re-warmup, 136 articles on re-warmup in soccer, and 154 articles on re-warmup during half-time. After the removal of duplicates and gray literature, after the screening and the eligibility assessments, a total of 26 articles met the inclusion criteria and were included in the present study (Figure 2). The literature demonstrates that passive recovery during halftime in intermittent team sports leads to a decrease in core temperature by 1.1 °C and muscle temperature by 2.0 °C,[10, 11] subsequently increasing the risk factors for injury.[2] It has been found that a short-duration, low-intensity warmup improves intermittent sprint performance after the interval.[12]

However, all studies reviewed in this investigation pointed out a decline in physical and cognitive performance when traditional resting times are observed during halftime under environmental conditions typically ranging from 10-30 °C. It should be noted that varying results have been observed under specific conditions, such as hot and cold environments, which have shown diverse muscle temperature responses. The results underscore the importance of re-warmup during both the post-warmup period and halftime break, as it helps maintain muscle temperature and flexibility[13] and enhances performance.[14] Specifically, re-warmup facilitates the ability to perform maximal sprints with incomplete recovery periods and jumping,[14] ultimately reducing the risk factors for injury during the match. In relation to the objective of our research, we identified the study by Yanaoka *et al.*,[15] in which it was shown that a 1-minute re-warmup at speed corresponding to 90% of VO_{2max} could prevent reductions in sprint and strength performance while increasing muscle temperature. Implementing a structured re-warmup of short duration can have practical applicability in races and can be integrated with other specific ergogenic strategies used during pre-race warmup, post-race recovery, and halftime[13] (Table I).

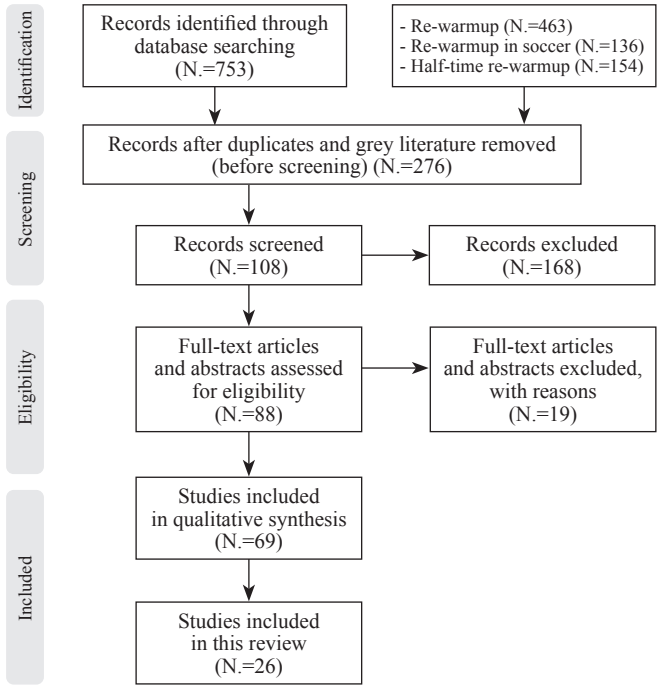

Figure 2.—Flow diagram of study design.

Discussion

In soccer, as in other team sports, it is imperative that warmups improve acute explosive performance. However, the exact strategies, methods, and consequences of different warmup practices remain unclear. An interval of time between warmup and the game and during halftime negates the positive metabolic effects of warmup. The lack of balance between scientific reality and practical applicability remains the major obstacle to overcome. A 3- to 7-min warmup (RW) produces physiological and performance benefits.[15, 17, 18] However, after passive time, a time-efficient and feasible RW protocol applicable in team sports is needed. This brief review aims to search the literature to see if there is any work at <1.30 s min that combines evidence and applicability.

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TABLE I.—*Highlights.*

Pre-competition warmup increases muscle temperature and enhances performance by reducing injury risk factors. However, the duration for which these benefits remain effective is still unclear[16]
On average, muscle temperature increases by 0.1 °C per minute during a 20-minute warmup period and subsequently decreases by the same amount per minute during 20 minutes of inactivity after the warmup[13]
A period of nine minutes of inactivity leads to a significant reduction of 3.8±0.6% in Countermovement Jump (CMJ) performance (P=0.046). Furthermore, a 23-minute period of inactivity is associated with a notable reduction of 7.3±0.7% in lay-up points (P=0.027)[13]
The re-warmup is a viable strategy for enhancing performance and reducing injury risk factors during extended periods of passive time. However, due to the limited methodological quality of the available studies, further exploration of these correlations is necessary, particularly in official or simulated competition settings

Passive times in soccer games

Some intermittent sports, including soccer, involve two consecutive periods of play lasting approximately 30-45 minutes, with a 10-20-minute interval separating them. Official pre-game protocols often include a passive resting period of around 10-15 minutes between the warmup and the start of the game, as well as between the first and second half (half-time). These passive periods have the potential to nullify the positive metabolic effects of the warmup, thereby increasing the risk factors for injury, particularly in the lower limb.[3, 19] Hence, this discussion aims to explore the practical applicability of re-warmup during the periods between the warmup and the start of the match, and during the half-time break of the match.

Re-warmup and before the match

The pre-match warmup routine in soccer aims to improve performance and prevent injuries. Soccer players commonly engage in a warmup session lasting for 30 minutes that includes a combination of low or high-intensity exercises, stretching, joint mobility, and technical drills involving the ball. This type of training has been proven to increase muscle temperature and induce physiological changes that are associated with enhanced sprinting and jumping performance, as well as decreased injury risk factors.[4, 8, 16]

Traditionally, after finishing the warmup, soccer players are allowed to return to the locker room. During this time, they may focus on technical and tactical aspects of the match and receive interventions from the medical staff. In international competitions (*i.e.*, Champions League), the teams' presentation and the playing of official anthems create a passive rest period of around 15-20 minutes before the match begins.

However, it is essential to note that the short-term impacts of the warmup, which heavily rely on changes in muscle temperature, may lessen during this passive rest period.[20] The inclusion of a re-warmup before the match remains a complex issue, as the manipulation of volume, intensity, and recovery can either positively or negatively affect subsequent performance.[17]

A study by Abade *et al.*[17] examined different re-warmup protocols (*i.e.*, none, eccentric, plyometric, and with repeated changes of direction) performed for six minutes following a standardized warmup on four consecutive days in a controlled crossover approach. Vertical jump and sprint performance were evaluated immediately and 12 minutes following the warmup. The outcomes revealed that repeated changes of direction and plyometric exercises had beneficial effects on jumping and sprinting performance. On the other hand, the eccentric protocol demonstrated potential negative implications, as it resulted in a reduction in the vertical jump, suggesting a possible detrimental effect on subsequent performance. This exploratory study[17] highlights the potential drawbacks of incorporating eccentric exercise before a soccer match. However, plyometric exercises and repeated changes of direction appear to be simple, quick, and effective activities that can help mitigate the decline in vertical jump and sprint ability after the warmup.

Additionally, a study has reported that muscle temperature increases by 0.1 °C per minute during warmup and decreases at the same rate during inactivity.[13] The decline in muscle temperature during a nine-minute period of inactivity was found to be associated with a 3.8±0.6% reduction in countermovement jump (CMJ) performance.

Other studies have shown that a 20-minute pre-competition warmup routine can increase muscle temperature, but the benefits may diminish following a period of inactivity, leading to a decrease in certain aspects of athletic performance.[13, 17, 21] Tsurubami *et al.*[4] examined the effects of warmup intensity and duration on jump performance. The results indicated that high and moderate-intensity warmups can maintain increased muscle temperature for up to 20 minutes. Jumping performance after a high-intensity warmup remained elevated for 20 minutes compared to a moderate-intensity warmup. It is recommended to perform a high-intensity warmup (80% VO_2max) if there is a rest period of more than 10 minutes between the warmup and the ath-

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letic event. A moderate-intensity warmup (60% VO_2max) is recommended[4] if the competition immediately follows the warmup. A relatively high-intensity warmup is necessary to sustain jump performance after a prolonged warmup and maintain muscle temperature for an extended duration.[4]

Re-warmup in half-time

The structure of a soccer game involves two consecutive periods of play (lasting ~30-45 minutes) that are separated by an interval of 10-20 minutes. Half-time practices employed by team sports players generally include rehydration and supplzementation strategies, resolving injury-related problems, receiving tactical instructions, and feedback from the coach. However, the typically passive nature of these actions has been associated with a decrease in cognitive[22] and performance abilities.[18] Although a re-warmup of 3 to 7 minutes yields physiological and performance benefits after the half-time interval,[15] a time-efficient and feasible protocol is needed for using a re-warmup in athletic settings.

Yanaoka *et al.*[12] studied the half-time re-warmup. The protocol consisted of 15 minutes of seated rest (control), 3 minutes of moderate-intensity re-warmup (cycling at 60% of maximal oxygen uptake [$\dot{V}o_2$max]), and 3 minutes of low-intensity re-warmup (cycling at 30% of $\dot{V}o_2$max). The Cycling Intermittent-Sprint Protocol consisted of 10 seconds of rest, 5 seconds of maximal sprinting, and 105 seconds of active recovery at 50% of $\dot{V}o_2$max, with repeated cycles for 20 minutes. This study revealed that players may increase their intermittent sprint performance by using a re-warmup for a very short duration at low and moderate intensities. These results support its use in athletics because the very short duration of re-warmup could allow it to be combined with other specific ergogenic strategies during the interval. Furthermore, this study suggested that a low-intensity re-warmup has the same rate of performance enhancement and less anaerobic energy expenditure than a moderate-intensity re-warmup. Therefore, according to these findings,[12] the short-duration, low-intensity re-warmup can be recommended for players of intermittent team sports.

A study by Bang and Park[18] points out that a re-warmup of 7 minutes during half-time confirmed that the decremental effect of static rest on sprint performance during the initial period of the second half could be mitigated by a re-warmup protocol at half-time. This study compared the effects of a 7-minute shuttle run during halftime with a control (rest) condition on subsequent athletic performance and lower leg temperature in the second half. Eighteen male soccer players (age 22, 179 cm, 70 kg, 10-year athletic career) randomly performed a 20 m shuttle run (at an intensity of 70% of maximum heart rate) and a seated rest (sitting on a bench) during halftime in two separate sessions. A 5-minute soccer simulation protocol, which consisted of soccer-specific activities (jumping, sprinting, kicking, passing, and dribbling at various intensities and distances), was repeated nine times to simulate the first and second half of a soccer match. The results of the study[18] confirmed that a decremental effect of static rest on sprint performance during the early period of the second half could be mitigated by a re-warmup of 7 minutes at half-time.[18] The author recommends moderate-intensity aerobic exercise (70% of maximum heart rate) during the interval to avoid the negative effects associated with interval sedentariness.

A study by Edholm *et al.*[19] studied two male professional players who performed traditional passive rest and low-intensity warmup during half-time of two soccer games. Sprinting and jumping performance were reduced by 2.6% and 7.6%, respectively, during half-time in the passive rest athlete, while sprinting performance was maintained, and the decline in jumping performance (3.1%) was less after re-warmup.

Christaras *et al.*[23] recently investigated the effect of a short warmup strategy in half-time re-warmup on the performance of young soccer players in physical ability and running tests during games. Twenty-three players (under 17 years of age) participated in the study. Body temperature, 10 m sprint, 30 m sprint, Illinois Agility Test (IAT), Countermovement Jump (CMJ), and Squat Jump (SJ) were measured immediately after warmup. After passive rest, a significant decrease in body temperature and performance in SJ was observed ($P<0.001$ for both). No other differences were observed. In conclusion, the short re-warmup program may limit the decrease in performance in power tests, such as sprints and jumps.

A study by Zois *et al.*[24] investigated the effects of a high-intensity, short-duration re-warmup on team sport-related performance in a randomized, crossover study. The re-warmup with five repetitions maximum leg-press improved physical performance. Small-Sided Game (SSG) improved skill execution after standardized intermittent exercise. Although a 3- to 7-minute re-warmup produces physiological and performance benefits after half-time, a scientifically efficient and applicable re-warmup protocol is needed and feasible in a unit of time.

Fashioni *et al.*[25] compared a 15-minute passive recovery strategy (CON) to a 3-minute re-warmup (RWU) in

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amateur soccer players. The 3-min RWU consisted of 2 sets (40 m) of jogging, 2 sets (40 m) of skipping, and one set (20 m) of dynamic stretch exercises followed by 2 sets each of straight sprinting, tuck jump into sprints and reaction sprints. The RWU group performed better in the 20-m sprint, squat jump, and CMJ height than the control group, highlighting how 3 minutes of re-warmup can positively affect soccer performance.

Mohr *et al.*[21] conducted a study where a 7-minute re-warmup at moderate intensity was compared to very low-intensity physical activities. The athletes were Danish soccer players competing in the Danish fourth division. The 7' of RWU consisted of running and other moderate-intensity exercises that corresponded at ~135bpm (or 70% of the peak heart rate reached in the game). Both core and muscle temperature were maintained, and sprint performance did not deteriorate after the half-time period.

An investigation comparing passive recovery to 5 minutes of intermittent agility exercises (IAE), and 5 minutes of intermittent whole-body vibration (WBV) was conducted by Lovell *et al.*[26] The IAE protocol involved soccer-specific runs (20 m) using the SAFT agility course, with the rest tailored to reach 70% of the maximal heart rate. In contrast, the WBV protocol provided for 3 sets of 60 seconds, where the players had to keep a partial-isometric-squat position (knee flexion of 30 degrees) on a platform that produced vertical sinusoidal vibration. Both conditions maintained the hamstrings' eccentric dynamic force, representing a valid opportunity to decrease the injury risk of these muscles. The IAE protocol resulted superior to the WBV for sprint performance, power, and dynamic strength performance, and to keep muscle temperature higher, although WBV did not deteriorate sprint and CMJ performance.

Lovell *et al.*[27] investigated the use of passive heating (PH), non-specific active heating (NSAH), and soccer-specific active heating (SSAH), comparing these different re-warmup strategies to a control group that performed a passive recovery (CON). During the PH trial, subjects were immersed in water at 40°C up to the gluteal fold. The NSAH procedure involved the athletes cycling on an ergometer at an intensity that elicits 70% of the maximal heart rate. In contrast, the SSAH procedure provided for sprint drills with rest manipulated to reach 70% of the maximal heart rate. The results highlighted that the NSAH and SSAH groups improved performance compared to the control group. The passive heating strategy was found to be ineffective in reducing performance decrement associated with the half-time rest period in elite young male soccer players.

Yanaoka *et al.*[15] compared fully passive rest and a 1-minute re-warmup during half-time to investigate the effect of a 1-minute high-intensity re-warmup on performance and physiological responses during the Loughborough Intermittent Shuttle Test (LIST). In a randomized, counterbalanced crossover design, 12 amateur players (soccer, basketball, handball, and lacrosse; age, 22±2 years; height, 1.70±0.08 m; body mass, 65.1±8.3 kg; Body Mass Index, 22.4±1.9 kg/m^2; VO_{2max}, 53.5±4.5 mL/kg/min) performed the LIST. The LIST included two 45-minute halves separated by a 15-minute half-time. Each half included repetitions of exercise cycles consisting of 3×20 m walking, 1×20 m maximal sprinting, 3×20 m jogging, and 3×20 m running. During half-time, participants were assigned to either a control (15 min seated rest) or a re-warmup group (1 min running at 90% maximal oxygen uptake after 14 min seated rest). Compared with the control group, re-warmup prevented a decrease in sprint performance at the fourth and sixth periods of the LIST (fourth: 2.4%, P=0.002, d=1.68, sixth: 3.6%, P=0.012, d=1.74). In addition, re-warmup reduced the amplitude of maximal voluntary contraction (MVC) electromyogram after HT (12%, P=0.017, d=1.12) without a decrease in maximal voluntary contraction force, suggesting an increase in neuromuscular efficiency (9%, P=0.048, d=0.58). Re-warmup also increased the mean heart rate in the early part of the second half (4 bpm, P=0.016, d=0.38). In conclusion, re-warmup improved sprint performance, core temperature, muscle activation, and heart rate in the second half of the LIST. The results suggest that re-warmup should be recommended for intermittent team sports players when longer re-warmups are not possible.[15]

This technical report revealed that soccer players could improve their sprint performance by using a 1-min high-intensity re-warmup. This result would support the idea that a re-warmup of short duration can be easily applied in official competitions and adapted to specific ergogenic strategies during half-time. It is possible that a re-warmup involving running between two lines 20 m apart, such as the one used in the study by Yanaoka,[15] could be performed on the field after leaving the locker room during half-time.

Towlson's study[28] highlights how 2.6 minutes are usually available for the re-warmup, showing that strategies that require that amount of time could be implemented, such as the protocol described by Yanaoka *et al.*,[15] positively affecting athletes' performance.

Practical applications

This investigation identified that 1-minute-re-warmup, performed at high intensity, appears to be a strategy that

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improves sprint performance, core temperature and muscle activation, combining a positive biofeedback between practical applicability and scientific evidence. This theoretical/practical reality finds application in passive rest, pre-game and half time of a soccer game and allows for adaptation to other specific ergogenic strategies. This allows coaches, and athletic trainers to adopt re-warmup in both professional and amateur soccer.

It is evident, however, that traditionally, in practical soccer reality, the coach and the players do not well accept re-warmup exercises. Official pre-match protocols and those in half time allow players to return to the locker room and perform, since time immemorial, routine activities such as rehydration strategies, clothing change, medical care, and technical and tactical instructions with even individual feedback from the coach and other technical staff figures. In contrast we need to further highlight how the exact strategies, methods and consequences of different re-warmup practices still remain insufficiently described in the literature. Thus, the existing studies lack complete methodological quality, as they are not related to a quantitative number of players. These relationships need to be further explored in official or simulated matches. The lack of balance between scientific reality and practical applicability remains the major obstacle to overcome. Currently, most of the time in passive rest is devoted to technical/tactical discussion and possible medical or physiotherapy practices. Our results provide essential information that soccer coaches can use to improve their teams' performance during matches.

Limitations of the study

This technical report has been limited to a concise narrative description of the main data obtained from the studies considered it did not evaluate the quality of studies as typical of systematic reviews.

The limited number of available studies prevents us from doing consensus evaluations on re-warmup. More high-quality studies with homogeneous study designs could help clarify the potential benefits of re-warmup in the passive phases of a soccer match. It is important to point out that, the re-warmup time is often insufficient due to a greater focus on technical reports.

Conclusions

This investigation identified that a high-intensity re-warmup lasting 1 minute improves sprint performance, core temperature, and muscle activation, combining practical applicability with scientific evidence; this is furthermore associated with a decreased risk of injury after passive rest, pre-competition warmup, and at half-time. These findings support the use of re-warmup in soccer, as a short-duration re-warmup can be easily applied during passive rest periods and integrated with other specific ergogenic strategies, particularly during half-time. The practicality of the conclusions drawn from this report allows re-warmup to be adopted in non-professional and youth soccer as well since no unique or sophisticated equipment is required. However, the validation and implementation of these findings in professional soccer require further extensive work and involvement of athletes.

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Conflicts of interest
The authors certify that there is no conflict of interest with any financial organization regarding the material discussed in the manuscript.

Authors' contributions
All authors read and approved the final version of the manuscript.

History
Article first published online: June 11, 2024. - Manuscript accepted: April 23, 2024. - Manuscript revised: March 25, 2024. - Manuscript received: December 30, 2023.